Version 7.4.7

PVsyst - Simulation report

Standalone system

Project: NEB Third Floor

Variant: New simulation variant

Standalone system with batteries

System power: 9.60 kWp

NEB Third Floor - Ghana

Author

PVsyst V7.4.7
VC0, Simulation date:
03/28/25 10:55
with V7.4.7

Project summary

Geographical Site
NEB Third Floor
Ghana

Situation

Latitude	6.67 °N
Longitude	-1.56 °W
Altitude	257 m
Time zone	UTC

Project settings

Albedo	0.20

Weather data
NEB Third Floor
Meteonorm 8.1, Sat=100% - Synthetic

System summary

Standalone system

Standalone system with batteries

PV Field Orientation

Fixed plane	
Tilt/Azimuth	9 / 0 °

User's needs
Daily household consumers
Constant over the year

Average	32.8 kWh/Day

System information

PV Array

Nb. of modules	32 units
Pnom total	9.60 kWp

Battery pack

Technology	Lithium-ion, LFP
Nb. of units	6 units
Voltage	48 V
Capacity	1200 Ah

Results summary

Useful energy from solar	11687.42 kWh/year	Specific production	1217 kWh/kWp/year	Perf. Ratio PR	61.14 %
Missing Energy	280.01 kWh/year	Available solar energy	15353.80 kWh/year	Solar Fraction SF	97.66 %
Excess (unused)	3404.51 kWh/year				

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General parameters

Standalone system | **Standalone system with batteries**

PV Field Orientation

Orientation		**Sheds configuration**	**Models used**	
Fixed plane		No 3D scene defined	Transposition	Perez
Tilt/Azimuth	9 / 0 °		Diffuse	Perez, Meteonorm
			Circumsolar	separate

User's needs

Daily household consumers
Constant over the year
Average 32.8 kWh/Day

PV Array Characteristics

PV module

Manufacturer	E.ON Energie Deutschland
Model	AURA FM300
(Original PVsyst database)	
Unit Nom. Power	300 Wp
Number of PV modules	32 units
Nominal (STC)	9.60 kWp
Modules	4 string x 8 In series
At operating cond. (50°C)	
Pmpp	8.66 kWp
U mpp	237 V
I mpp	37 A

Battery

Manufacturer	Huawei
Model	Luna2000-10-SO, with inverter 5 kW
Technology	Lithium-ion, LFP
Nb. of units	6 in parallel
Discharging min. SOC	10.0 %
Stored energy	51.8 kWh
Battery Pack Characteristics	
Voltage	48 V
Nominal Capacity	1200 Ah (C10)
Temperature	Given monthly values

Controller

Universal controller

Technology	MPPT converter
Temp coeff.	-5.0 mV/°C/Elem.
Converter	
Maxi and EURO efficiencies	97.0 / 95.0 %

Battery Management control

Threshold commands as	SOC calculation
Charging	SOC = 0.96 / 0.80
Discharging	SOC = 0.10 / 0.35

Total PV power

Nominal (STC)	10 kWp
Total	32 modules
Module area	52.4 m²
Cell area	45.6 m²

Array losses

Thermal Loss factor

Module temperature according to irradiance

Uc (const)	20.0 W/m²K
Uv (wind)	0.0 W/m²K/m/s

DC wiring losses

Global array res.	108 mΩ
Loss Fraction	1.5 % at STC

Serie Diode Loss

Voltage drop	0.7 V
Loss Fraction	0.3 % at STC

Module Quality Loss

Loss Fraction -0.8 %

Module mismatch losses

Loss Fraction 2.0 % at MPP

Strings Mismatch loss

Loss Fraction 0.1 %

IAM loss factor

Incidence effect (IAM): Fresnel smooth glass, n = 1.526

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.998	0.981	0.948	0.862	0.776	0.636	0.403	0.000

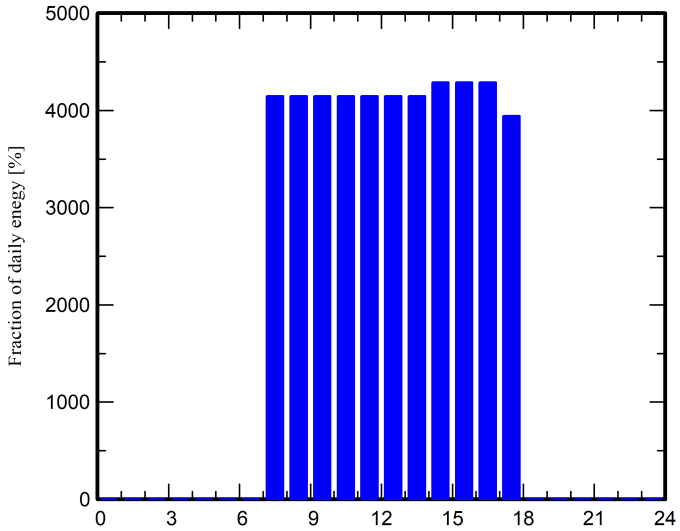
Detailed User's needs

Daily household consumers, Constant over the year, average = 32.8 kWh/day

Annual values

Use 5 days a week	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
Lamps(LED), Camera	79	31/lamp	11.0	27156
TV	2	110/app	11.0	2420
Ceiling Fans	14	44/app	11.0	6776
Floodlights	4		4	560
Projectors	1		10	3440
Digital Podium	1	344 tot	11.0	3784
Camera, Laptop, Smoke detectors	1	156 tot	11.0	1716
Total daily energy				45852

Hourly distribution



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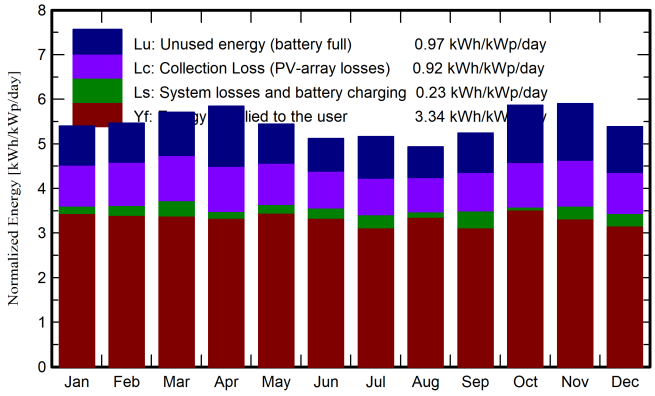
Main results

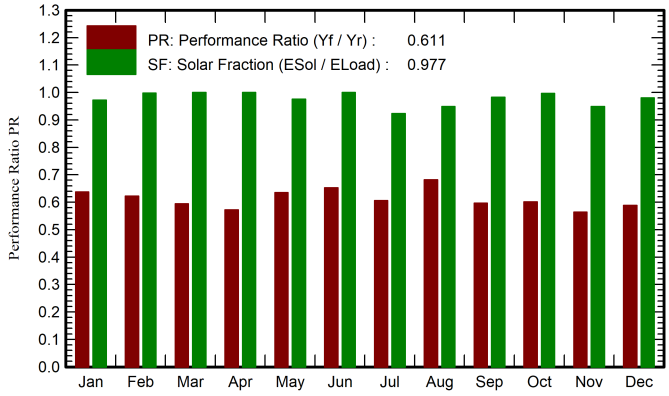
System Production

Useful energy from solar	11687.42 kWh/year	Perf. Ratio PR	61.14 %
Available solar energy	15353.80 kWh/year	Solar Fraction SF	97.66 %
Excess (unused)	3404.51 kWh/year		
Loss of Load		**Battery aging (State of Wear)**	
Time Fraction	0.0 %	Cycles SOW	98.4 %
Missing Energy	280.01 kWh/year	Static SOW	90.0 %

Normalized productions (per installed kWp)



Performance Ratio PR



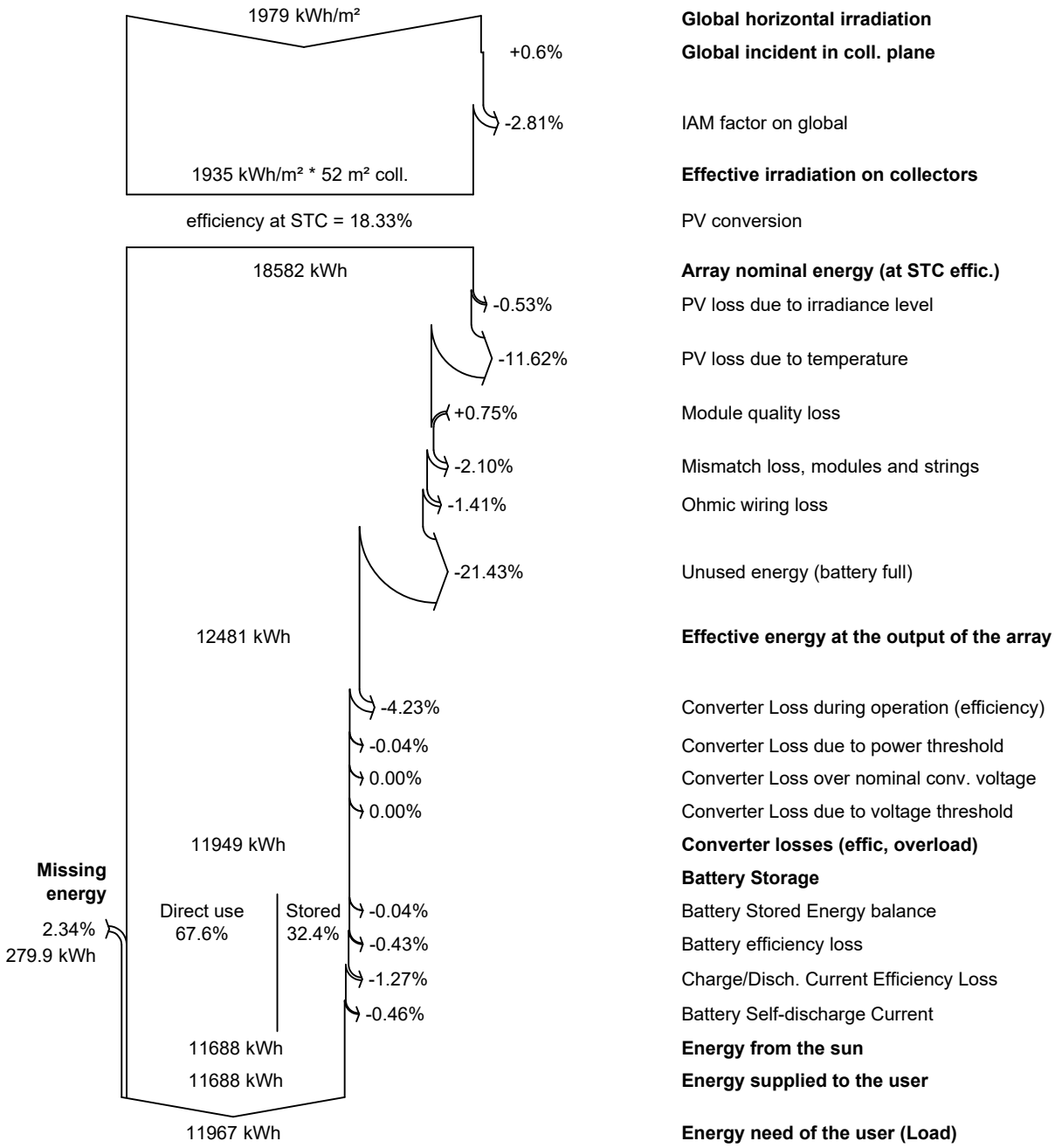
Balances and main results

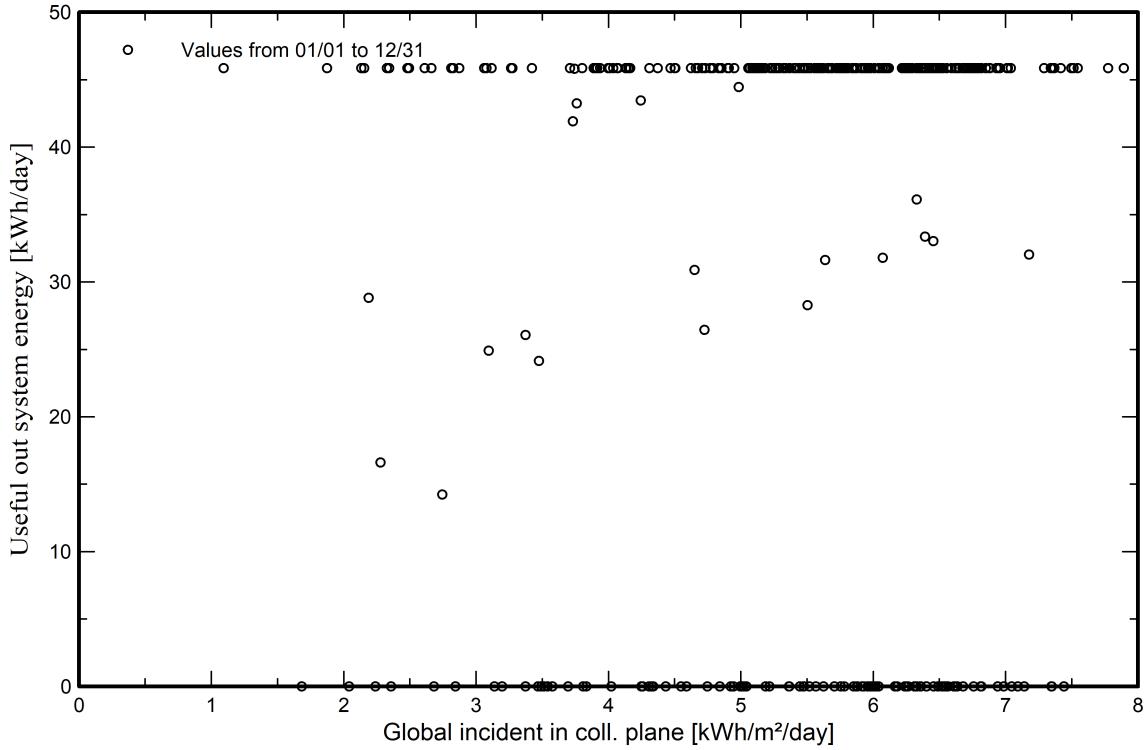
	GlobHor	GlobEff	E_Avail	EUnused	E_Miss	E_User	E_Load	SolFrac
	kWh/m²	kWh/m²	kWh	kWh	kWh	kWh	kWh	ratio
January	157.8	163.0	1291	258.1	29.21	1025	1055	0.972
February	148.0	149.3	1168	233.9	2.40	915	917	0.997
March	175.0	172.4	1351	286.1	0.00	1009	1009	1.000
April	178.5	170.8	1348	386.1	0.05	963	963	1.000
May	176.4	163.1	1299	258.7	25.36	1029	1055	0.976
June	162.7	148.1	1193	207.9	0.00	963	963	1.000
July	168.2	154.8	1246	274.8	77.90	931	1009	0.923
August	157.0	148.2	1192	201.8	54.01	1001	1055	0.949
September	157.0	152.8	1217	251.0	16.46	901	917	0.982
October	176.1	177.1	1402	379.3	3.94	1051	1055	0.996
November	166.1	172.7	1361	363.6	51.26	957	1009	0.949
December	155.8	162.8	1286	303.1	19.41	943	963	0.980
Year	1978.7	1935.1	15354	3404.5	280.01	11687	11967	0.977

Legends

GlobHor	Global horizontal irradiation	E_User	Energy supplied to the user
GlobEff	Effective Global, corr. for IAM and shadings	E_Load	Energy need of the user (Load)
E_Avail	Available Solar Energy	SolFrac	Solar fraction (EUsed / ELoad)
EUnused	Unused energy (battery full)		
E_Miss	Missing energy		

Loss diagram



Predef. graphs

Daily Input/Output diagram